

Problem-Solution Fit

Date	10 JULY 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) (CS) • Students preparing for exams • College and university learners • Online learning platform users • Teachers and mentors • Educational institutions • Individuals seeking emotional well-being support • Researchers in AI and NLP	6. CUSTOMER LIMITATIONS (CL) • Limited awareness of AI tools. • Limited access to mental health resources. • Difficulty expressing emotions. • Lack of personalized educational support. • Privacy concerns regarding emotional data. • Limited technical knowledge.	5. AVAILABLE SOLUTIONS (ASI) Pros • Chatbots provide basic conversation. • Emotion detection APIs are available. • Mental health applications offer support. • Machine learning models provide accurate emotion classification. Cons • Generic responses with little personalization. • Most systems don't combine emotion detection with learning support. • Limited educational recommendations. • Many solutions require multiple platforms. • Few systems use hybrid AI models.
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2. PROBLEMS / PAINS + ITS FREQUENCY (PR)	9. PROBLEM ROOT / CAUSE [RC]	7.BEHAVIOR+ITS INTENSITY (BE)
• Difficulty identifying emotions from text. • Lack of personalized learning guidance. • Students often experience stress and anxiety while studying. • Existing systems provide generic responses. • Emotional issues negatively affect learning performance. • Frequent need for instant emotional assessment and support.	• Students experience academic stress. • Traditional learning systems ignore emotional state. • Lack of real-time emotion analysis. • Generic learning recommendations. • Limited integration of AI with education. • ☐ Emotional well-being is often overlooked.	• Frequently searches for study assistance. • Uses AI tools during learning. • Wants quick and accurate responses. • Needs emotional support during stressful situations. • Uses educational applications regularly. • High demand during examination periods.
3. TRIGGERS TO ACT (TR) • Feeling stressed before exams. • Difficulty concentrating while studying. • Need for emotional support during learning. • Want personalized study recommendations. • Need quick emotion detection from text. • Improve learning efficiency through AI assistance.	10. YOUR SOLUTION (SL) Develop an AI-powered Emotion Detection & Learning Support Engine that combines BiLSTM and fine-tuned BERT models to accurately detect emotions from user text. The system classifies emotions such as Joy, Sadness, Anger, Fear, Love, Surprise, and Neutral, and generates personalized learning recommendations using the Google Gemini API. The application is built with Streamlit, providing an intuitive interface for users to receive instant emotion analysis, AI-generated learning support, and	8. CHANNELS OF BEHAVIOR (CH) Online • Streamlit Web Application • GitHub Repository • Educational websites • Online learning platforms • Google Gemini API Offline • Colleges and Universities • Classrooms • Study groups • Academic workshops • Research laboratories

4. EMOTIONS BEFORE / AFTER (EM)

Before

- Stressed
- Confused
- Demotivated
- Frustrated
- Emotionally overwhelmed

After

- Relaxed
- Motivated
- Confident
- Better focused
- Supported with personalized guidance

emotion-aware educational guidance in real time.